XISOSIS · Institutional Research Paper

TOWARDS A CLASSIFICATION OF THE FREE GROUP FACTORS $L_{\mathbb{K}}(F_k)$ OVER $\mathbb{R}$, $\mathbb{C}$ AND $\mathbb{H}$ AS REAL VON NEUMANN ALGEBRAS

Main result: $M_2(L(F_k)) \cong L(F_{(k+3)/4})$, so the quaternionic factor $L_{\mathbb{H}}(F_5)$ and the real factor $L_{\mathbb{R}}(F_2)$ have the same complexification $L(F_2)$; complexification therefore cannot decide their real isomorphism class, which remains open.

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Abstract

This paper gives a complete solution of Exercise 6 of the Cooperman–Petrunina Challenge Series and a corrected account of the results that surround it. Using the Dykema–Rădulescu amplification formula we show $M_2(L(F_k)) \cong L(F_{(k+3)/4})$, and deduce that the quaternionic free group factor $L_{\mathbb{H}}(F_5)$ and the real free group factor $L_{\mathbb{R}}(F_2)$ have the same complexification, $L(F_2)$. We also prove $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$ (Exercise 5) and relate it to the sign of J_n^2, give a self-contained proof of the finite-parameter part of the Dykema–Rădulescu dichotomy, and fix the Jordan–Wigner degree count for $\mathbb{Q}$-structured generators.

Scope. These results do not solve Problem 2. Problem 2 asks for a classification of $L_{\mathbb{K}}(F_k)$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, and in particular for decisions on $L(F_2) \cong L(F_3)$ and on $L_{\mathbb{H}}(F_5) \cong L_{\mathbb{R}}(F_2)$. Both questions remain open after this paper. What the paper establishes is why complexification cannot decide the second question (Section 4), and which claimed invariants in the earlier draft do not hold (Appendix A).

1. $\mathbb{Q}$-structure and Brauer parity

Throughout, K is complex conjugation on $(\mathbb{C}^2)^{\otimes n}$, $J_1 = \sigma_y K$ and $J_n = \sigma_y^{\otimes n} K$. The map φ of [2] identifies $\mathbb{Q} \cong \mathbb{H}$ with the real algebra $\{M \in M_2(\mathbb{C}) : J_1 M J_1^{-1} = M\}$, and φ extends to an isomorphism $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$.

Lemma 1.1 (parity of J_n^2). $J_n^2 = (-1)^n I$.

Proof. Since $KAK = \bar{A}$ for any matrix A,

$$J_n^2 = \sigma_y^{\otimes n} K \sigma_y^{\otimes n} K = \sigma_y^{\otimes n}\, \bar{\sigma}_y^{\otimes n}.$$

The matrix σ_y is purely imaginary, so $\bar{\sigma}_y = -\sigma_y$, and

$$J_n^2 = (-1)^n \left(\sigma_y^2\right)^{\otimes n} = (-1)^n I. \qquad \blacksquare$$

Proposition 1.2 (the symmetry algebra $\mathfrak{k}_n$). Let $\mathfrak{k}_n = \{A \in \mathfrak{su}(2^n) : J_n A J_n^{-1} = A\}$.

1. For a Pauli string P of weight $w(P)$, $J_n P J_n^{-1} = (-1)^{w(P)} P$, so $iP \in \mathfrak{k}_n$ if and only if $w(P)$ is odd.
2. $\mathfrak{k}_n \cong \mathfrak{sp}(2^{n-1})$ for odd n and $\mathfrak{k}_n \cong \mathfrak{so}(2^n)$ for even n, and

$$\dim \mathfrak{k}_n = \sum_{w \text{ odd}} \binom{n}{w} 3^w = \frac{4^n - (-2)^n}{2}.$$

Proof. (1) For one qubit, $\sigma_y \bar{\sigma}_\alpha \sigma_y = -\sigma_\alpha$ for $\alpha \in \{x, y, z\}$ and $\sigma_y \bar{I} \sigma_y = I$; tensoring gives $J_n P J_n^{-1} = (-1)^{w(P)} P$. Because J_n is antilinear, $J_n(iP)J_n^{-1} = -i(-1)^{w(P)}P$, which equals iP exactly when $w(P)$ is odd. The strings $\{iP\}$ form a basis of $\mathfrak{su}(2^n)$, and conjugation by J_n is diagonal in that basis, so $\mathfrak{k}_n$ is spanned by the odd-weight strings.

(2) The unitaries commuting with an antiunitary J on $\mathbb{C}^N$ form the compact symplectic group $Sp(N/2)$ when $J^2 = -I$ and a conjugate of $O(N)$ when $J^2 = +I$ [12]. Lemma 1.1 with $N = 2^n$ gives the identification. The count of odd-weight strings is $\frac{1}{2}[(1+3)^n - (1-3)^n]$, which equals $\dim \mathfrak{sp}(2^{n-1}) = 2^{n-1}(2^n+1)$ for odd n and $\dim \mathfrak{so}(2^n) = 2^{n-1}(2^n-1)$ for even n. $\blacksquare$

n	J_n^2	$\mathfrak{k}_n$	$\dim \mathfrak{k}_n = (4^n - (-2)^n)/2$
1	$-I$	$\mathfrak{sp}(1) = \mathfrak{su}(2)$	3
2	$+I$	$\mathfrak{so}(4)$	6

n	J_n^2	$\mathfrak{k}_n$	$\dim \mathfrak{k}_n = (4^n - (-2)^n)/2$
3	$-I$	$\mathfrak{sp}(4)$	36
4	$+I$	$\mathfrak{so}(16)$	120

Lemma 1.3 (Exercise 5: $\mathbb{H}\otimes_{\mathbb{R}}\mathbb{H} \cong M_4(\mathbb{R})$). The $\mathbb{R}$-linear map $\psi : \mathbb{H}\otimes_{\mathbb{R}}\mathbb{H} \to \mathrm{End}_{\mathbb{R}}(\mathbb{H}) \cong M_4(\mathbb{R})$ defined on elementary tensors by

$$\psi(a \otimes b)(x) = a\, x\, \bar{b}, \qquad a, b, x \in \mathbb{H},$$

is an isomorphism of real algebras.

Proof. Homomorphism: since quaternionic conjugation reverses products, $\overline{b_1 b_2} = \bar{b}_2 \bar{b}_1$, and

$$\psi(a_1 a_2 \otimes b_1 b_2)(x) = a_1 a_2\, x\, \bar{b}_2 \bar{b}_1 = \psi(a_1 \otimes b_1)(\psi(a_2 \otimes b_2)(x)).$$

Injectivity: the kernel of ψ is a two-sided ideal of $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H}$, which is central simple because it is a tensor product of central simple $\mathbb{R}$-algebras. The kernel is proper because $\psi(1 \otimes 1) = I$, so it is $\{0\}$. Surjectivity: both algebras have real dimension 16, and an injective linear map between spaces of equal finite dimension is bijective. ■

Corollary 1.4 (order of $[\mathbb{H}]$ and the sign of J_n^2). The class $[\mathbb{H}]$ has order exactly 2 in $\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2$. Moreover, the real algebra $\mathcal{C}_n = \{M \in M_{2^n}(\mathbb{C}) : J_n M J_n^{-1} = M\}$ satisfies

$$\mathcal{C}_n \cong \mathbb{H}^{\otimes_{\mathbb{R}} n} \cong \begin{cases} M_{2^{n-1}}(\mathbb{H}), & n \text{ odd}, \\ M_{2^n}(\mathbb{R}), & n \text{ even}, \end{cases}$$

so $[\mathcal{C}_n] = [\mathbb{H}]^n$, and this class is non-trivial exactly when $J_n^2 = -I$.

Proof. Lemma 1.3 gives $[\mathbb{H}]^2 = [M_4(\mathbb{R})] = 0$, and $[\mathbb{H}] \neq 0$ because $\mathbb{H}$ is a division algebra other than $\mathbb{R}$. For $\mathcal{C}_n$: conjugation by $J_n = J_1^{\otimes n}$ acts factor by factor, so $\varphi(\mathbb{H})^{\otimes n} \subseteq \mathcal{C}_n$. The left side has real dimension 4^n. The fixed points of an antilinear involution on $M_{2^n}(\mathbb{C})$ form a real form, also of real dimension 4^n, so the inclusion is an equality. Iterating Lemma 1.3 gives the two cases, and Lemma 1.1 matches them with the sign of J_n^2. ■

2. Majorana degree and the size of the dynamical Lie algebra

We use the Jordan–Wigner map with Z on site:

$$\gamma_{2j-1} = \Big(\prod_{k<j} Z_k\Big) X_j, \qquad \gamma_{2j} = \Big(\prod_{k<j} Z_k\Big) Y_j, \qquad \{\gamma_r, \gamma_s\} = 2\delta_{rs} I.$$

Then

$$Z_j = -i\gamma_{2j-1}\gamma_{2j}, \qquad X_j X_{j+1} = -i\gamma_{2j}\gamma_{2j+1}.$$

Lemma 2.1 (degree of $\mathbb{Q}$-structured generators).

1. $Z_j Z_{j+1} Z_{j+2} = i\,\gamma_{2j-1}\gamma_{2j}\gamma_{2j+1}\gamma_{2j+2}\gamma_{2j+3}\gamma_{2j+4}$, a Majorana monomial of degree 6.
2. A single-site X_j or Y_j is not quadratic: $X_j = (\prod_{k<j} Z_k)\gamma_{2j-1}$ is a monomial of odd degree $2j-1$.

Proof. (1) Multiply the three on-site expressions; the factors involve disjoint indices, so no cancellation occurs, and $(-i)^3 = i$. (2) Each Z_k with $k < j$ contributes two Majoranas and γ_{2j-1} contributes one. ■

Lemma 2.2 (quadratic closure). For distinct r, s, t,

$$[\gamma_r\gamma_s,\, \gamma_s\gamma_t] = 2\,\gamma_r\gamma_t.$$

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Hence the real span of $\{\gamma_r\gamma_s : r < s\}$ is a Lie algebra isomorphic to $\mathfrak{so}(2n)$, of dimension $n(2n-1)$. A circuit whose generators are all quadratic monomials has a dynamical Lie algebra (DLA) of dimension $O(n^2)$. For product-state inputs, its expectation values, gradients and output probabilities can be computed classically in poly(n) time [6, 18, 26, 27].

Proof. $\gamma_r\gamma_s\gamma_s\gamma_t = \gamma_r\gamma_t$. Moving γ_s from the right of $\gamma_s\gamma_t\gamma_r\gamma_s$ to the left anticommutes it past γ_r and γ_t, so $\gamma_s\gamma_t\gamma_r\gamma_s = \gamma_t\gamma_r = -\gamma_r\gamma_t$. Subtracting gives $2\gamma_r\gamma_t$. Brackets of quadratic monomials on overlapping index pairs are quadratic, and those on disjoint pairs commute, so the span is closed. Its dimension is $\binom{2n}{2} = n(2n-1)$. ■

Remark 2.3 (what degree does and does not decide). A generator of degree greater than 2 leaves the free-fermion algebra $\mathfrak{so}(2n)$, but it does not by itself make the DLA exponential, and an exponential DLA is not known to imply classical hardness.

- The commuting strings $\{Z_jZ_{j+1}Z_{j+2}\}_{j=1}^{n-2}$ are degree 6, yet they generate an abelian DLA of dimension $n-2$, and the circuit is trivially sampleable from $|0\cdots0\rangle$ (Exercise 7 of [1]).
- For the entangling ansatz $\{X_j, Z_j\}\cup\{Z_jZ_{j+1}Z_{j+2}\}$, the DLA equals $\mathfrak{k}_n$, of dimension $\Theta(4^n)$. This is verified by exact Pauli closure for $n \le 9$ in [2] and for $n \le 8$ by the validation scripts (Appendix B); for general n it is a conjecture.
- Whether polynomial DLA dimension characterises efficient classical sampling within this class is Problem 3 of the series and is open.

Proposition 2.4 (cluster ansatz, from [2], Theorem 3). In the convention with X on site, the generators $\{X_j\}\cup\{Z_jX_{j+1}Z_{j+2}\}$ map to quadratic monomials along two disjoint paths of $2\lceil n/2\rceil$ and $2\lfloor n/2\rfloor$ Majoranas. Quadratic monomials along a connected tree on m Majoranas generate $\mathfrak{so}(m)$, by Lemma 2.2. Therefore

$$\mathfrak{g}_{\mathrm{cl}} \cong \mathfrak{so}(2\lceil n/2\rceil) \oplus \mathfrak{so}(2\lfloor n/2\rfloor) \subseteq \mathfrak{k}_n.$$

The inclusion in $\mathfrak{k}_n$ holds because these particular generators have odd weight (1 or 3). It is not a property of quadratic monomials in general: X_jX_{j+1} is quadratic but has weight 2, so it is not in $\mathfrak{k}_n$.

3. Interpolated free group factors

For a II_1 factor M with trace τ and $t > 0$, the amplification M_t is pMp with $\tau(p) = t$ when $t \le 1$, and $p\,M_d(M)\,p$ with $(\tau\otimes\mathrm{Tr})(p) = t$ for any integer $d \ge t$ when $t > 1$. It is well defined up to isomorphism, $M_d(M) \cong M_d$ for integers d, and the fundamental group is $\mathcal{F}(M) = \{t > 0 : M_t \cong M\}$.

Theorem 3.1 (amplification formula; Dykema [19], Rădulescu [20]). For $1 < r \le \infty$ and $t > 0$,

$$L(F_r)_t \cong L\Big(F_{1+\frac{r-1}{t^2}}\Big).$$

This is a cited theorem; we do not reprove it. For integers $n, k \ge 2$ it extends Voiculescu's compression formula $L(F_n)_{1/k} \cong L(F_{1+k^2(n-1)})$, proved with semicircular and circular matrix models. Dykema and Rădulescu define $L(F_r)$ for non-integer r so that the formula holds for all t. We also use Dykema's free product formula $L(F_r) * L(F_s) \cong L(F_{r+s})$ for $1 < r, s \le \infty$ [19].

Theorem 3.2 (dichotomy). Exactly one of the following holds.

1. $L(F_r) \cong L(F_s)$ for all $1 < r, s \le \infty$; equivalently $L(F_2) \cong L(F_3)$. In this case $\mathcal{F}(L(F_r)) = \mathbb{R}_+^*$ for every r.
2. $L(F_r) \cong L(F_s)$ only if $r = s$. In this case $\mathcal{F}(L(F_r)) = \{1\}$ for $1 < r < \infty$.

Proof for finite parameters. Suppose $L(F_x) \cong L(F_y)$ with $1 < x < y < \infty$, and put $d = y - x$.

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Step 1 (translation). For every $u > 1$, the free product formula gives

$$L(F_{x+u}) \cong L(F_x) * L(F_u) \cong L(F_y) * L(F_u) \cong L(F_{y+u}).$$

So $L(F_z) \cong L(F_{z+d})$ for every $z > x + 1$.

Step 2 (scaling). Amplifying both sides by $t > 0$ and applying Theorem 3.1 gives $L(F_w) \cong L(F_{w+d/t^2})$ for every $w > 1 + x/t^2$. Writing $\varepsilon = d/t^2$, translation by any $\varepsilon > 0$ preserves the isomorphism class on parameters $w > 1 + (x/d)\,\varepsilon$.

Step 3 (chaining). Let $1 < a < b < \infty$. Choose an integer N so large that $\varepsilon = (b-a)/N$ satisfies $1 + (x/d)\varepsilon < a$. Every point of $a, a+\varepsilon, \ldots, b$ then lies above the threshold, so

$$L(F_a) \cong L(F_{a+\varepsilon}) \cong \cdots \cong L(F_b).$$

If instead $L(F_x) \cong L(F_\infty)$ with x finite, then for $u > 1$ we get $L(F_{x+u}) \cong L(F_\infty) * L(F_u) \cong L(F_\infty) \cong L(F_x)$, which reduces to the finite case.

Fundamental groups: for finite r, $t \in \mathcal{F}(L(F_r))$ means $L(F_{1+(r-1)/t^2}) \cong L(F_r)$. In alternative 2 this forces $t = 1$. In alternative 1 it holds for every $t > 0$.

That the collapse of all finite parameters also includes $r = \infty$ is part of the cited theorem [19, 20]; we do not reprove it. Since the two alternatives are exhaustive, alternative 1 holds if and only if $L(F_2) \cong L(F_3)$. ■

4. Exercise 6 and the complexification boundary

Definitions. $L_{\mathbb{R}}(F_k)$ is the real group von Neumann algebra generated by the left regular representation of F_k on $\ell^2_{\mathbb{R}}(F_k)$, so that $L_{\mathbb{R}}(F_k) \otimes_{\mathbb{R}} \mathbb{C} \cong L(F_k)$. The quaternionic free group factor is

$$L_{\mathbb{H}}(F_k) := L_{\mathbb{R}}(F_k) \otimes_{\mathbb{R}} \mathbb{H}.$$

The challenge statement [1] is authoritative for these definitions. This is the definition under which the complexification stated in Exercise 6 holds.

Lemma 4.1 (Exercise 6, first part). For $1 < k \leq \infty$,

$$M_2(L(F_k)) \cong L\Big(F_{(k+3)/4}\Big).$$

Proof. $M_2(L(F_k)) \cong L(F_k)_2$ by the definition of amplification. Theorem 3.1 with $r = k$, $t = 2$ gives $L(F_k)_2 \cong L(F_{1+(k-1)/4})$, and

$$1 + \frac{k-1}{4} = \frac{k+3}{4}. \qquad \blacksquare$$

For $k \geq 2$ the new parameter satisfies $(k+3)/4 \geq 5/4 > 1$, so it lies in the admissible range; $k = \infty$ gives $L(F_\infty)$.

Proposition 4.2 (Exercise 6, second part). $L_{\mathbb{H}}(F_5) \otimes_{\mathbb{R}} \mathbb{C} \;\cong\; L(F_2) \;\cong\; L_{\mathbb{R}}(F_2) \otimes_{\mathbb{R}} \mathbb{C}$.

Proof. Using $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$ (Section 1),

$$L_{\mathbb{H}}(F_5) \otimes_{\mathbb{R}} \mathbb{C} \cong \big(L_{\mathbb{R}}(F_5) \otimes_{\mathbb{R}} \mathbb{C}\big) \otimes_{\mathbb{C}} \big(\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C}\big) \cong L(F_5) \otimes_{\mathbb{C}} M_2(\mathbb{C}) \cong M_2(L(F_5)).$$

Lemma 4.1 with $k = 5$ gives $M_2(L(F_5)) \cong L(F_{8/4}) = L(F_2)$. Finally $L_{\mathbb{R}}(F_2) \otimes_{\mathbb{R}} \mathbb{C} \cong L(F_2)$ by definition. ■

Theorem 4.3 (complexification does not reflect isomorphism).

1. There are real von Neumann algebras $A \not\cong B$ with $A \otimes_{\mathbb{R}} \mathbb{C} \cong B \otimes_{\mathbb{R}} \mathbb{C}$.
2. For finite-dimensional central simple real algebras, $A \cong B$ if and only if $\dim A = \dim B$ and $[A] = [B]$ in $\mathrm{Br}(\mathbb{R})$.
3. For the real hyperfinite II_1 factor $\mathcal{R}_{\mathbb{R}} = \overline{\bigotimes}_{m\geq 1} M_4(\mathbb{R})$, we have $\mathcal{R}_{\mathbb{H}} := \mathcal{R}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{H} \cong \mathcal{R}_{\mathbb{R}}$, so the Brauer parity cancels.
4. Whether $L_{\mathbb{H}}(F_5) \cong L_{\mathbb{R}}(F_2)$ is not decided by complexification. By Proposition 4.2 their complexifications coincide, and this question is open.

Proof. (1) Take $A = \mathbb{H}$ and $B = M_2(\mathbb{R})$. $\mathbb{H}$ is a division algebra and $M_2(\mathbb{R})$ has zero divisors, so $A \not\cong B$, while both complexify to $M_2(\mathbb{C})$.

(2) This is Wedderburn's theorem: a central simple $\mathbb{R}$-algebra is $M_m(D)$ with $D \in \{\mathbb{R}, \mathbb{H}\}$ determined by its Brauer class, and m is then fixed by the dimension.

(3) By Lemma 1.3, $M_4(\mathbb{R}) \cong \mathbb{H} \otimes \mathbb{H}$. Regrouping the trace-preserving infinite tensor product,

$$\bigotimes_{m\geq 1} (\mathbb{H} \otimes \mathbb{H}) = \mathbb{H} \otimes \bigotimes_{m\geq 1} (\mathbb{H} \otimes \mathbb{H}),$$

which gives a trace-preserving isomorphism of the algebraic tensor products. It extends to the weak closures, so $\mathcal{R}_{\mathbb{R}} \cong \mathbb{H} \otimes \mathcal{R}_{\mathbb{R}} = \mathcal{R}_{\mathbb{H}}$.

(4) This follows from Proposition 4.2 and part (1). No argument in this paper, or in the earlier draft, decides the real isomorphism question. ■

Remark 4.4 (numerical evidence). Genesis script 2 of [1] reports finite-size corrections of about $+5/N$, 0 and $-2.5/N$ to the moments of the $\beta = 1, 2, 4$ random-matrix models, with identical limits. This illustrates that the large-N moment limit does not see the Brauer parity. It is a numerical observation, not a proof, and it does not bear on real isomorphism, because isomorphic moment limits are already implied by Proposition 4.2.

5. β-microstate free entropy dimension (corrected statement)

Let H_N^β denote the self-adjoint $N \times N$ matrices over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$ for $\beta = 1, 2, 4$. Their real dimension is

$$d_\beta(N) = N + \beta \frac{N(N-1)}{2}, \quad \text{that is,} \quad \frac{N(N+1)}{2}, \quad N^2, \quad N(2N-1).$$

For a self-adjoint tuple $X = (X_1, \dots, X_k)$, the microstate set $\Gamma_\beta(X; m, N, \varepsilon) \subset (H_N^\beta)^k$ consists of the tuples whose mixed moments of order at most m are within ε of those of X. The β-free entropy χ_β is defined as in [Voiculescu] with $\log \mathrm{vol}\, \Gamma_\beta$ normalised by $d_\beta(N)$ in place of N^2, together with the matching logarithmic correction. The modified dimension is

$$\delta_{0,\beta}(X) = k + \limsup_{\varepsilon \to 0} \frac{\chi_\beta(X + \varepsilon S : S)}{|\log \varepsilon|},$$

where S is a free semicircular family free from X.

Conjecture 5.1 (β-invariance). For every finite self-adjoint tuple X in a II_1 factor with Connes-embeddable moments,

$$\delta_{0,1}(X) = \delta_{0,2}(X) = \delta_{0,4}(X).$$

For a free semicircular family $S_1, \dots, S_k$, the common value would be k.

Status. The case $\beta = 2$ is Voiculescu's theory, and $\delta_{0,2}(S_1, \dots, S_k) = k$ is his theorem. Independent GOE or GSE matrices are asymptotically free, so the microstate sets Γ_1 and Γ_4 of a semicircular family are non-empty for large N. That fact does not compute their volume, and Conjecture 5.1 remains the open entry point listed under Problem 2 in [1].

Why the earlier "Theorem 1.2" is withdrawn.

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1. It asserted $\delta = k$ for every generating tuple. This is false already for $\beta = 2$. For free semicirculars S_1, S_2, the tuple (S_1, S_2, S_1) generates $L(F_2)$, but $\delta_0(S_1, S_2, S_1) = \delta_0(S_1, S_2) = 2 < 3$, since δ_0 is unchanged when an element of the generated $*$-algebra is adjoined.
2. Its key estimate $|\log \operatorname{vol} \Gamma_\beta - \log \operatorname{vol} \Gamma_2| = O(1/N)$ compares volumes in spaces of different real dimension ($d_1 \neq d_2 \neq d_4$). The two logarithms differ at order $N^2 \log N$, not $O(1/N)$.
3. The dimension list it used, $\dim \mathfrak{o}(N)$, $\dim \mathfrak{u}(N)$, $\dim \mathfrak{sp}(N/2)$, gives Lie algebras of compact groups. The microstate spaces are the self-adjoint spaces H_N^β with the dimensions $d_\beta(N)$ above.

6. Status of Problem 2

Question	Status after this paper
Exercise 5: $\mathbb{H} \otimes \mathbb{H} \cong M_4(\mathbb{R})$, order of $[\mathbb{H}]$, link to J_n^2	Proved (Lemma 1.3, Corollary 1.4)
Exercise 6: $M_2(L(F_k)) \cong L(F_{(k+3)/4})$; same complexification	Proved (Lemma 4.1, Proposition 4.2)
Dichotomy for $L(F_r)$	Known theorem [19, 20]; finite-parameter proof given (Theorem 3.2)
$L(F_2) \cong L(F_3)$?	Open
$L_{\mathbb{H}}(F_5) \cong L_{\mathbb{R}}(F_2)$?	Open; shown undecidable by complexification (Theorem 4.3)
Real and quaternionic dichotomy (entry point 1 of [1])	Open
β-invariance of free entropy dimension (entry point 2 of [1])	Open (Conjecture 5.1)
Classification of real forms of $L(F_k)$	Open

A solution of Problem 2 still requires an invariant of real von Neumann algebras that is finer than complexification and computable on free group factors, or an explicit real isomorphism.

Appendix A. Corrections relative to the chained draft

Draft result	Verdict	Correction
Part I, Lemma 1.1: JW map with $X_j = i\gamma_{2j-1}\gamma_{2j}$, then $Z_j = -i\gamma_{2j-1}\gamma_{2j}$	Inconsistent	Two JW conventions were mixed. With X on site, $Z_2Z_3Z_4 = -\gamma_1\gamma_2\gamma_3\gamma_6\gamma_7$ (degree 5, non-local). One convention is now fixed throughout (Lemma 2.1).
Part I, Lemma 1.1: "1-local Pauli rotations preserve quadratic constraints"	False	Only the on-site Pauli is quadratic; X_j has odd degree $2j-1$ (Lemma 2.1).
Part I, Lemma 1.1 and Part V, Lemma 2.1: degree 6 $\Rightarrow$ $\dim \mathfrak{g} \to O(4^n)$ $\Rightarrow$ "not in P"	False	Counterexample: commuting ZZZ strings, $\dim \mathfrak{g} = n - 2$. Exponential DLA is verified only for the entangling ansatz ($n \leq 9$); hardness is open (Remark 2.3).
Part V, Lemma 2.1: $[i\gamma_r\gamma_s, i\gamma_s\gamma_t] = 2i\gamma_r\gamma_t$	Wrong constant	The bracket equals $-2\gamma_r\gamma_t$; equivalently $[\gamma_r\gamma_s, \gamma_s\gamma_t] = 2\gamma_r\gamma_t$ (Lemma 2.2).

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Draft result	Verdict	Correction
Part I, Lemma 1.2: $J_n^2 = (-1)^n I$, sp/so alternation, dimensions	Correct	Missing step added: the odd-weight criterion and antilinearity (Proposition 1.2).
Part II, Lemma 1.1: $\mathbb{H} \otimes \mathbb{H} \cong M_4(\mathbb{R})$	Correct	Kernel-is-an-ideal step made explicit; Exercise 5's Brauer and J_n^2 parts added (Corollary 1.4).
Part II, Theorem 1.2: $\delta_1^* = \delta_2^* = \delta_4^* = k$	False as stated; proof invalid	Withdrawn; replaced by Conjecture 5.1 (Section 5).
Part III, Lemma 1.1: amplification formula	Statement correct; proof incorrect	M_t for $t > 1$ redefined correctly; the formula is cited as the Dykema–Rădulescu theorem, not derived from "free dimension metric" scaling (Theorem 3.1).
Part III, Theorem 1.2: dichotomy	Statement correct; proof incomplete	Proof via the free product formula, scaling and chaining (Theorem 3.2).
Part V, Theorem 1.1: choice $s = \sqrt{(t_1 - 1)/(t_2 - 1)}$	Wrong	With that s, $1 + (t_2 - 1)/s^2 \neq t_1$; the correct choice is $s = \sqrt{(t_2 - 1)/(t_1 - 1)}$. The claim "non-trivial $\Rightarrow \mathcal{F} = \mathbb{R}_+^*$" was unproved and is now proved (Theorem 3.2).
Part III, Lemma 1.3: $\mathrm{Isom}(...) \cong L(F_2)$; "non-isomorphic real forms"	Type error; unproved	A set of isomorphisms is not a factor; non-isomorphism of $L_{\mathbb{H}}(F_5)$ and $L_{\mathbb{R}}(F_2)$ is open (Theorem 4.3).
Part IV, Lemmas 6.1–6.2 and Part V, Lemma 1.2: Exercise 6	Correct	Definition of $L_{\mathbb{H}}(F_k)$ supplied; complexification computed (Lemma 4.1, Proposition 4.2).
Part IV, Theorem 6.3: proof via $L_{\mathbb{H}}(F_5)$, $L_{\mathbb{R}}(F_2)$ and the $\pm/N$ corrections	Statement true; proof invalid	It used an unproved non-isomorphism. Now proved with $\mathbb{H}$ versus $M_2(\mathbb{R})$; the hyperfinite cancellation is proved; the random-matrix corrections are relabelled as numerics (Theorem 4.3, Remark 4.4).
Front matter: "solution to Problem 2"	Overstated	Exercise 6 is solved; Problem 2 remains open (Section 6).

Appendix B. Numerical verification

The validation scripts (Python 3 and NumPy, MIT License) are in the GitHub repository listed on the front page; `python run_all.py` runs all 173 checks. All identities below were checked in exact arithmetic or to machine precision with NumPy (8- to 32-dimensional matrices).

Check	Result
$J_n^2 = (-1)^n I$, $n = 1, \ldots, 5$	Confirmed
Odd-weight count equals $(4^n - (-2)^n)/2$ and $\dim \mathfrak{sp}$ or $\dim \mathfrak{so}$, $n = 1, \ldots, 6$	3, 6, 36, 120, 528, 2016
ψ of Lemma 1.3 is a homomorphism and has rank 16	Confirmed

Check	Result
Majorana anticommutation; $Z_j = -i\gamma_{2j-1}\gamma_{2j}$; $X_jX_{j+1} = -i\gamma_{2j}\gamma_{2j+1}$; $Z_1Z_2Z_3 = i\gamma_1\cdots\gamma_6$	Confirmed ($n = 4$)
$[\gamma_r\gamma_s, \gamma_s\gamma_t] = 2\gamma_r\gamma_t$; draft's $2i\gamma_r\gamma_t$	Confirmed; draft value fails
Entangling ansatz DLA $= \dim \mathfrak{k}_n$, $n = 3, \ldots, 8$	36, 120, 528, 2016, 8256, 32640
Cluster DLA $= a(2a-1) + b(2b-1)$, $n = 3, \ldots, 7$	7, 12, 21, 30, 43
Commuting ZZZ DLA $= n - 2$, $n = 3, \ldots, 7$	1, 2, 3, 4, 5

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